

At erection peaks, where vehicles wait for the crane decides whether its queue reaches the whole yard

Cranes as dock resources, the p curve and its threshold read at each fleet size, crane starvation, H2's rise and fall, and a candidate dock-mouth proposition

Dock-mouth blocking at erection peaks: crane utilisation, waiting position and dock-road granularity

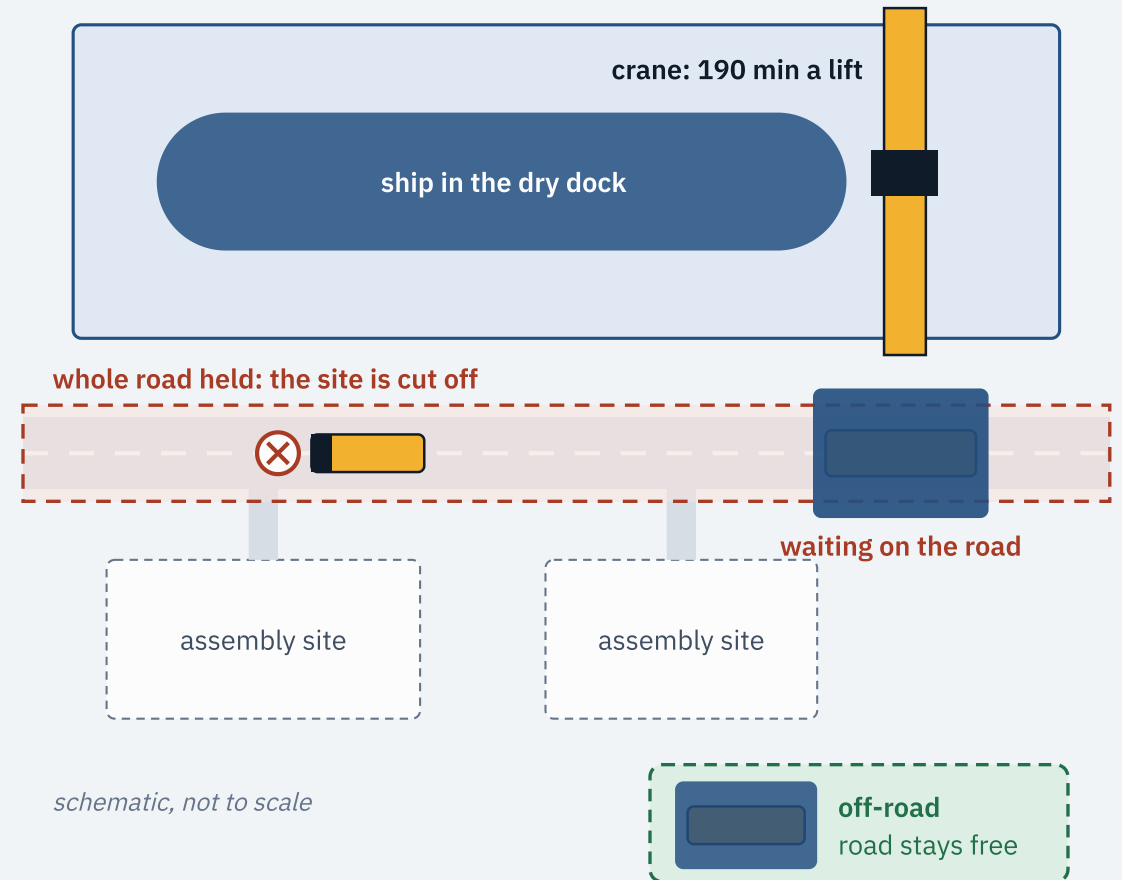
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Basis: thesis-core a882742 (T20–T23, reviewed)

Research manifesto v1 · the author's review (9 Oct 2026) · literature check T24

Dock mouth = crane + dock stopping road (in series)



This sub-study in one minute

1

The question

At erection peaks, when does the dock mouth (crane and dock stopping road in series) become the yard's bottleneck? The manifesto makes H1 part 3 conditional: it depends on whether vehicles waiting for the crane block the way to other stops.

2

What T22 reads

Whole-route reservation, waiting off-road, never falls on either yard, at either granularity, for $K = 20\text{--}150$ and p up to 0.99 (at most -1.7%). Rules that wait on the road, safe release without teleporting included, all fall with p ; the threshold moves left with K and right with splitting.

3

Effective capacity

At $p = 0.99$ safe release uses only 74–87% of crane-hours, reservation 96–97%: nominal $p < 1$, yet effective capacity falls short. The erection mix's capacity interval $[611, 703]$ becomes $[662, 1,241]$ with the road split.

4

Next

T27 (running): a less conservative deadlock-free rule that re-checks the interval's lower edge and H2's decline. Then T25: the three-resource network and the block-length curve; T26: floating docks and a finite holding area S . The author writes the proposition.

The reading in one sentence: whether a crane queue reaches the whole yard depends on where vehicles wait, how the dock road is split, and how many wait on it.

The order of this briefing (click to jump)

1

Question and position

Manifesto items served: H1 part 3, H2, the candidate proposition, H7

p. 4

2

Cranes and the p curve

Cranes and erection cadence; the p curve at each K; how the threshold moves with K and splitting; readings for H1 part 3

p. 5

3

Mechanism and effective capacity

Where vehicles wait; crane starvation; teleports counted by task and the split road; the erection mix's capacity interval

p. 9

4

When the decline appears (H2)

Safe release rises then falls; the no-clearing switch; three switches and T4

p. 13

5

The proposition and floating docks

Starvation caused by blocking; the three-resource network; floating docks and H7

p. 15

6

Progress, plan and journals

Established, running and queued tasks; target journals (school directory 2026)

p. 18

Question and position: the dock-mouth section of chapter 3, serving H1 part 3, H2, the candidate proposition and H7's floating-dock case

At erection peaks, through what channel does the crane queue become a capacity loss for the whole yard, and can the waiting position and the granularity of the dock road keep it at the dock mouth?

MANIFESTO ITEM	TEXT (RESEARCH MANIFESTO V1)	THIS STUDY SUPPLIES
H1, part 3	Whether the dock mouth becomes the yard's bottleneck depends on whether vehicles waiting for the crane block the way to other stops, that is, where they wait and whether the dock road is used in sections.	ρ curves at each K: three rules $\times$ two road granularities $\times K = 20\text{--}150$ (T22)
H2	As long as the extra vehicles occupy network resources, throughput falls beyond K_0 , and K_0 moves right with the holding-area capacity S.	Safe release rising then falling and the no-clearing switch (T22); re-check in T27; holding area S (T26); T4
Candidate	Dock-mouth proposition: with nominal load $\rho < 1$, starvation caused by blocking destabilises the system; with off-road waiting or a dock road used in sections, stability is restored.	The crane starvation rate (read in T22); the three-resource network (T25)
H7 case	Multi-transporter carriage of giant blocks for floating docks.	A floating-dock scenario (quay-side sink + floating crane, T26)

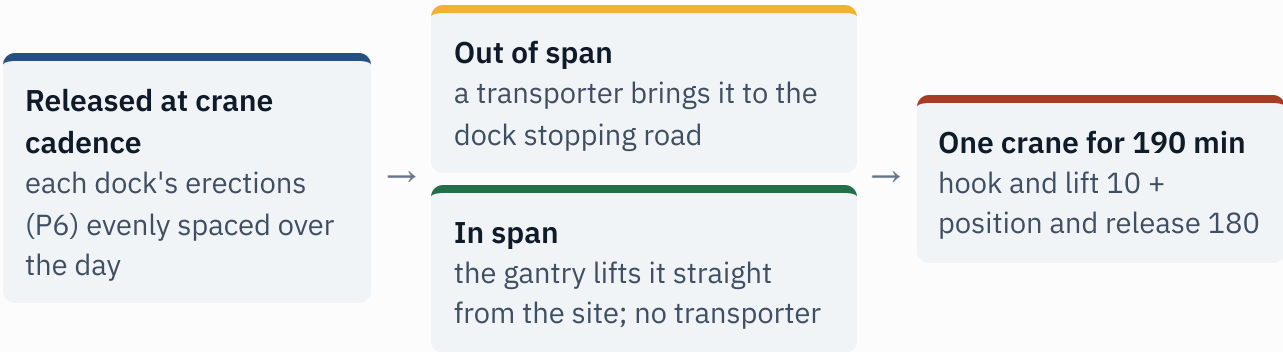
Related claims: C2's falling branch appears “when waiting crowds the road”; C3's orchestration includes waiting points and block granularity. Each conclusion states its conditions once: traffic rule, task-volume basis, resource granularity, yard and map version (manifesto §5).

Sources: research manifesto v1 §2 (C2, C3), §3 (H1, H2, H7), §4 (T4 and the candidate proposition), §5 (wording rules). The English and Korean of manifesto items are faithful translations; the Chinese governs.

Cranes are dock resources: erections are released at crane cadence, and in-span lifts take the crane but not the road

YARD · DOCK	CRANES COUNTED (CAPACITY ≥ 100 T)	NO.
Okpo · dock 1	900 t gantry, 200 t jib crane	2
Okpo · dock 2	900 t gantry	1
Yantai · dock 1	20,000 t bridge crane (Taishan), 370 t gantry	2
Yantai · dock 2	370 t gantry	1

Jib and tower cranes below 100 t (the mean block weight) are not counted; assigning Yantai's two 370 t gantries one per dock is a scenario choice. Okpo gantry spans: 206 m on dock 1 (Riviera 2014), 148 m on dock 2 (The Motorship).



$$\rho = P6/\text{day} \times 190 \text{ min} \div (n_{\text{crane}} \times 960 \text{ min})$$

Three cranes erect at most $3 \times 960 \div 190 = 15.2$ a day

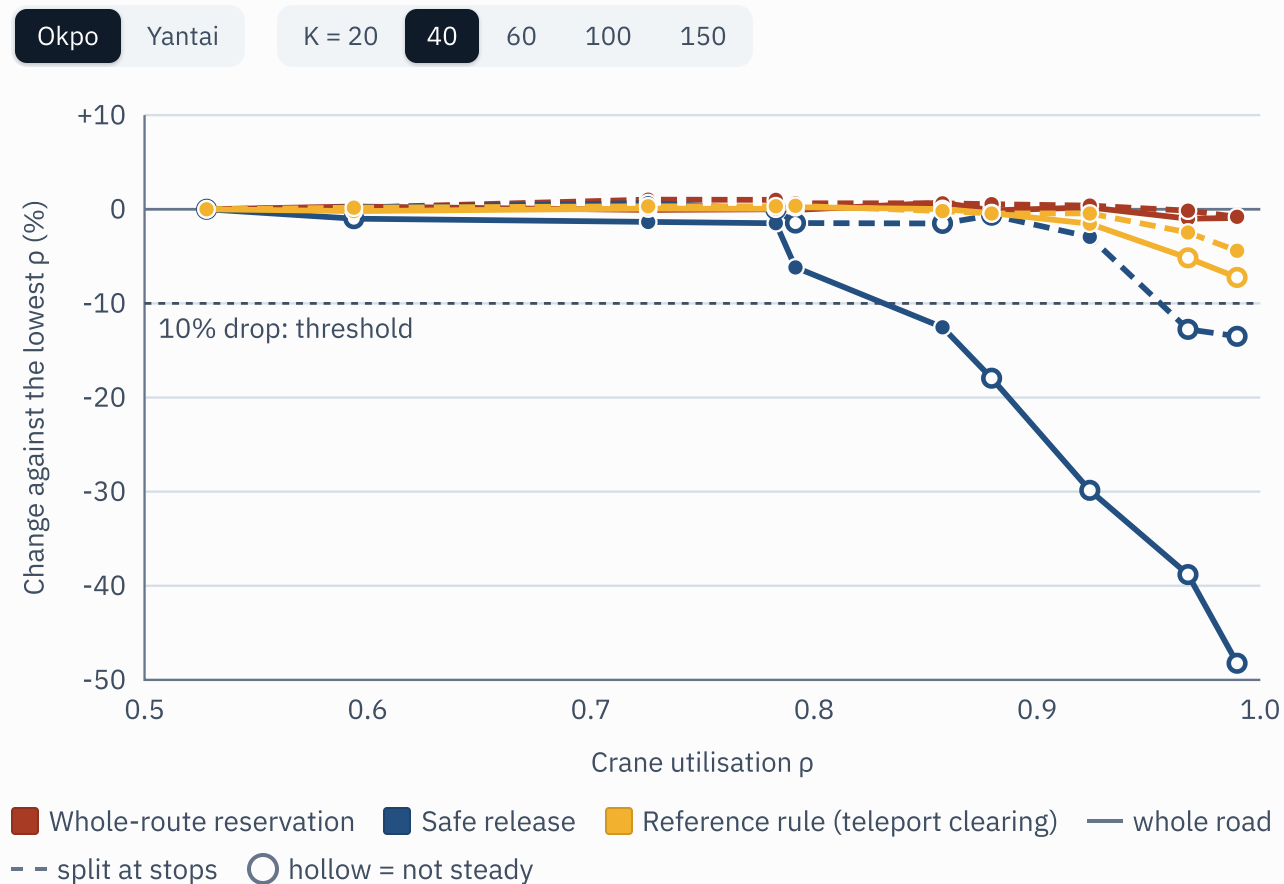
0.88 (0.78-0.97)
Okpo: $P6 = 2 \times 20 \text{ ships} \times 100 \div 300 = 13.3$ a day (range 11.9–14.7).
20 = LNG carriers delivered in 2025 (estimated from IR), 100 = midpoint of DSME's patent “usually 90–110”, 2 = peak day. **A scenario reading, not part of H1** (author, 9 Oct 2026).

0.13
Yantai: $P6 = 2$ a day from public output; the cranes are far from tight.

12.5%
Okpo's in-span share: zone 018 → dock 1 and zone 014 → dock 2 (span cover ≥ 0.5 on the map); none at Yantai.

Sources: thesis-core a882742 (T20 and T21 unchanged since 83c16bf), T20 report §1 (crane lists: Hanwha Ocean Facilities page; CIMC Raffles 2012 EIA notice; Hoist Magazine 2008), T21 report §2–3, t21_span.csv, t21_p6.csv. Crane cycle from ZhangSG2025, lines 2525–2528 (another yard); units per ship from DSME patent KR20090050364A and Roh & Lee 2006, p. 116 (89); ships a year estimated from Hanwha Ocean IR.

The ρ curve at each K: reservation (off-road) does not fall; on-road rules, safe release without teleports included, all fall with ρ



-0.9% / -0.8%

Reservation (waiting off-road): change at $K = 40$ and $p = 0.99$ (whole / split)

0.83 → 0.96*

Safe release (no teleporting), threshold at $K = 40$ (change at $p = 0.99$ if under 10%): whole → split

-7.2% → -4.4%

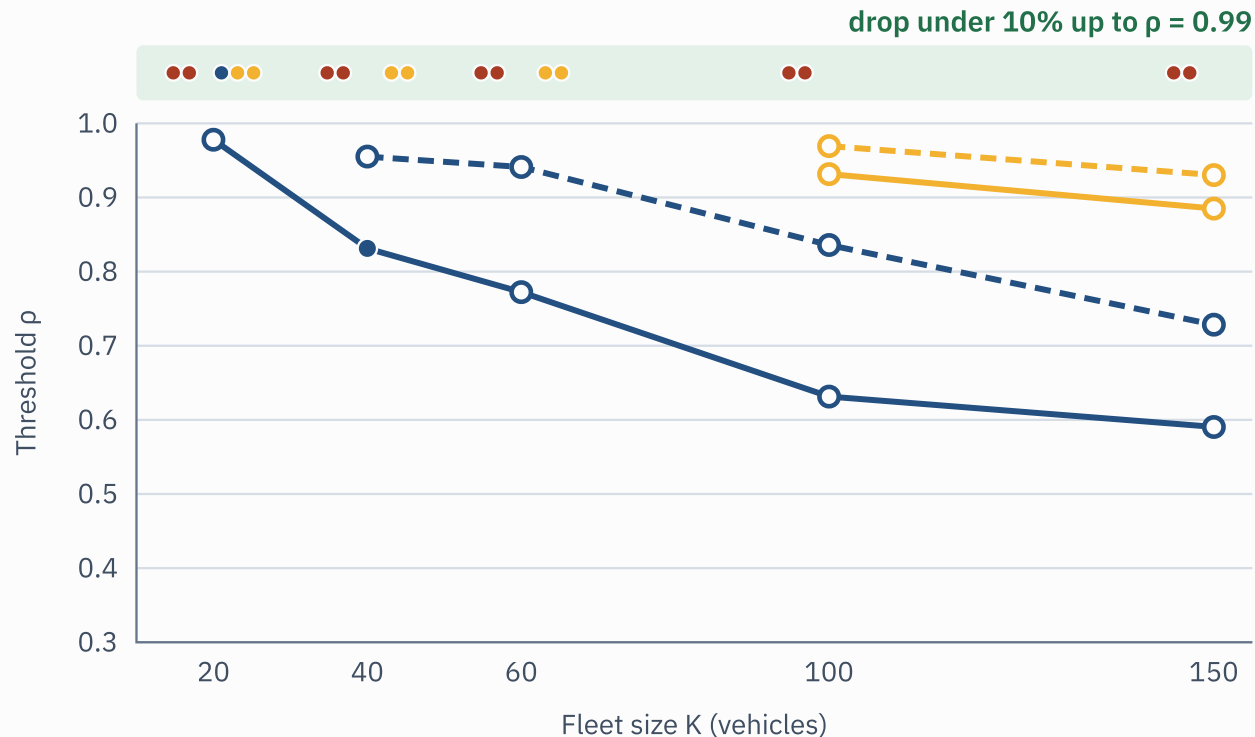
Reference rule (teleport clearing), threshold at $K = 40$ (change at $p = 0.99$ if under 10%): whole → split

Threshold = the p at which the drop first reaches 10%; * = it uses non-steady points. Safe release is already non-steady at the lowest p ; its slide at $K = 100-150$ comes from the criterion itself, not from the dock.

Sources: thesis-core a882742, results/t22_report_3287d62c7916/t22_rho_k.csv (T22 report §3.1); p batches 54f5e26030e2 (Okpo) and 16d6dde8b72d (Yantai), with reservation and the reference rule on the whole road reused from T21's 1b06c6d46f0d and 87d02d32bc13 (reproduced bit for bit). Conditions: the crane-cadence p levels, saturated demand, 16 h working day, 8 days (day 1 warm-up), 10 seeds; change against the yard's lowest p (Okpo 0.53, Yantai 0.13); hollow = fails the steady-state test (mix shift $\leq 2\%$, daily trend $\leq 5\%$). Maps: Okpo derived map rev 1825, Yantai r396; from T27's step zero on, maps come from 20_数据/船厂地图 (Okpo rev 1833), and T25 runs on them.

The threshold moves left with fleet size and right with splitting; reservation never falls: no single threshold fits every rule and K

Okpo Yantai



0.98* → 0.59*

Safe release, whole road: threshold from K = 20 to K = 150

0.83 → 0.96*

At K = 40, splitting the dock road at its stops moves the safe-release threshold right (whole → split)

0.89* → 0.93*

Reference rule (teleport clearing), K = 150: whole → split

-1.7%

Reservation: largest drop at $p = 0.99$ over both granularities and every K = 20–150

■ Whole-route reservation
 ■ Safe release
 ■ Reference rule (teleport clearing)
 — whole road
 - - split at stops
 ○ hollow = not steady

Sources: thesis-core a882742, t22_rho_k.csv and T22 report §3.1–3.2. Threshold = the p at which throughput first falls 10% below the lowest p (linear between neighbouring levels); hollow and * = the points behind the threshold include non-steady ones; green band at the top = the drop stays under 10% up to $p = 0.99$ (hover for the change at $p = 0.99$). Conditions as on the previous slide.

H1 part 3: T22 supplies readings that name the rule and K; the wording and the main basis are the author's to set after T27

H1 part 3 (manifesto v1): whether the dock mouth becomes the yard's bottleneck depends on whether vehicles waiting for the crane block the way to other stops, that is, where they wait and whether the dock road is used in sections.

Current condition note (author, 9 Oct 2026): the old thresholds (Okpo about 0.85, Yantai about 0.6–0.7) were read only under the reference rule (teleport clearing) with the dock road as one resource. T22 shows the threshold moves with rule, granularity and K; the table gives every cell.

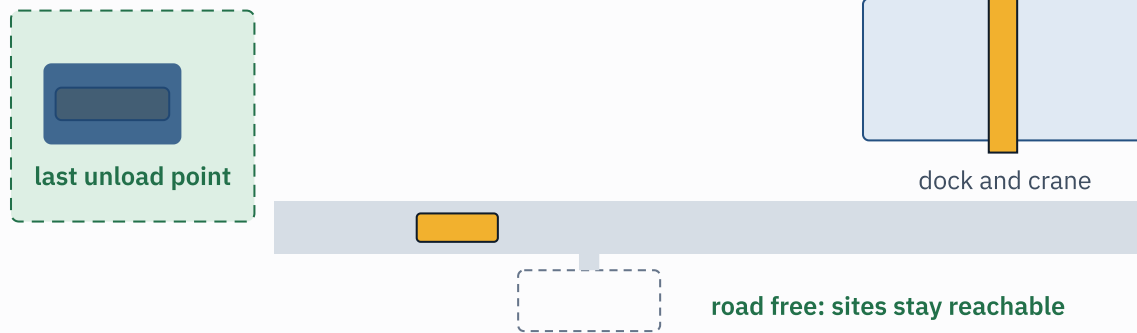
RULE · GRANULARITY	OKPO					YANTAI				
	K = 20	K = 40	K = 60	K = 100	K = 150	K = 20	K = 40	K = 60	K = 100	K = 150
■ Whole-route reservation · whole road	– (–1.2)	– (–0.9)	– (–0.8)	– (–1.7)	– (–0.6)	– (+0.5)	– (+1.8)	– (+0.9)	– (+1.2)	– (+1.4)
■ Whole-route reservation · split at stops	– (–1.6)	– (–0.8)	– (–1.2)	– (–0.2)	– (–0.8)	– (+1.4)	– (+0.8)	– (+1.8)	– (+0.7)	– (+1.8)
■ Safe release · whole road	0.98*	0.83	0.77*	0.63*	0.59*	0.93*	0.70*	0.56	0.39*	0.38*
■ Safe release · split at stops	– (–4.5)	0.96*	0.94*	0.84*	0.73*	0.97*	0.80*	0.70	0.55*	0.43*
■ Reference rule (teleport clearing) · whole road	– (–3.7)	– (–7.2)	– (–9.8)	0.93*	0.89*	0.98*	0.90*	0.85*	0.79*	0.66
■ Reference rule (teleport clearing) · split at stops	– (–2.2)	– (–4.4)	– (–6.4)	0.97*	0.93*	– (–5.7)	0.94*	0.88*	0.80*	0.74*

For the author, after T27: the wording of H1 part 3 (the report's “qualitative layer plus a numeric layer naming rule and K” is one option); the new numbers of H1 part 2; the main basis (16 h or 24 h working day, daily mean or peak). If T27 changes the best deadlock-free rule, the safe-release rows gain a less conservative companion.

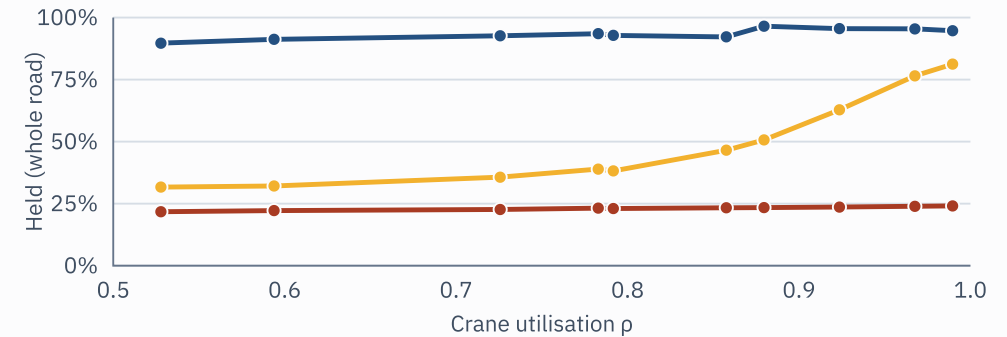
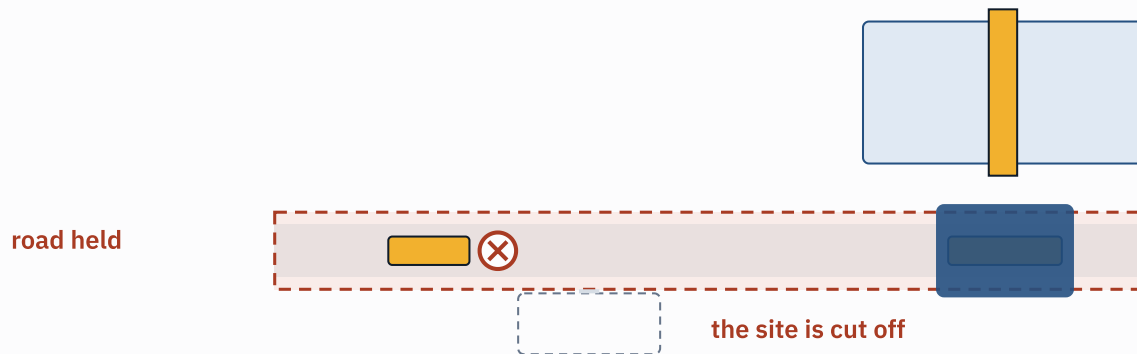
Sources: research manifesto v1 §3, H1 (quoted); thesis-core a882742, t22_rho_k.csv and T22 report §1.3, §3.1 and §15; reviews/T22.md §5 (three items for the author). Bold = threshold p; * = uses non-steady points; – (x) = drop under 10% up to p = 0.99, change at p = 0.99 (%) in brackets. Conditions: saturated demand, 16 h working day, 8 days, 10 seeds, Okpo rev 1825, Yantai r396.

The mechanism: the difference is where vehicles wait – at their last unloading point (off-road) or on the dock road

Reservation: crane slot booked with the route



Waiting on the road (safe release, reference rule)



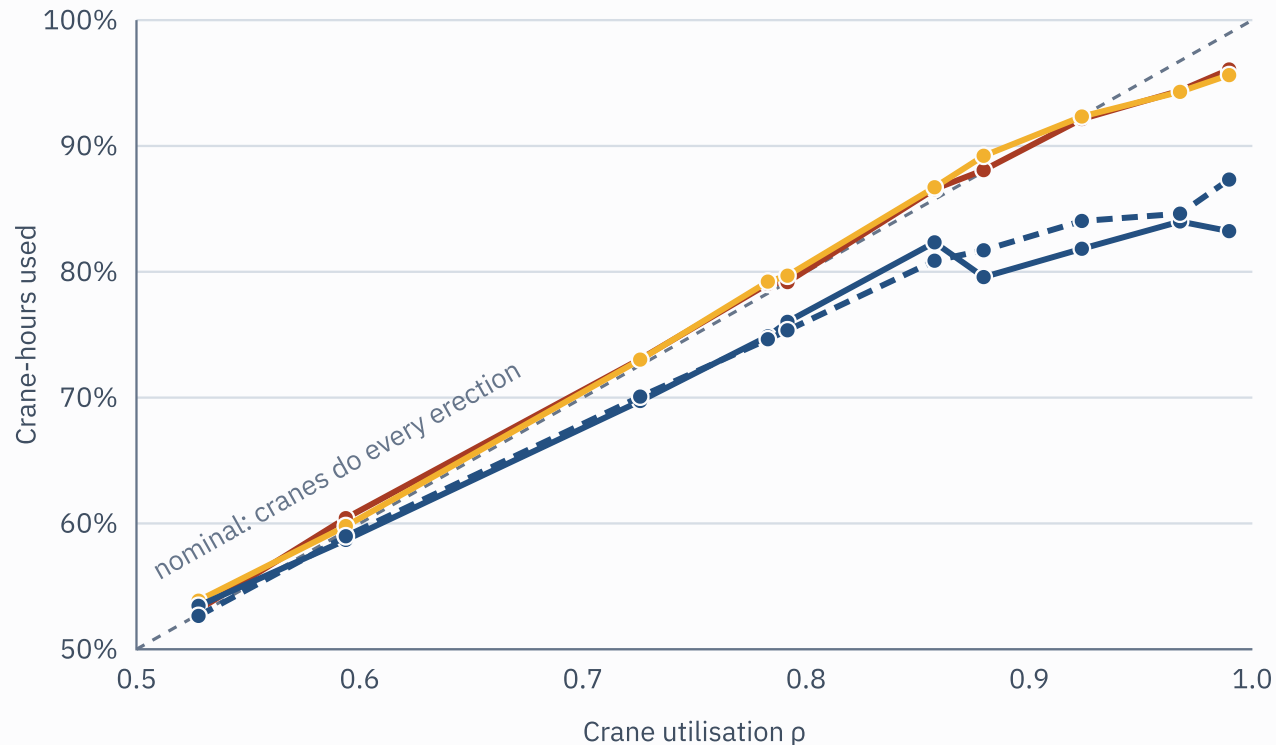
■ Whole-route reservation ■ Safe release ■ Reference rule (teleport clearing)

- **The cost moves off-road:** at $p = 0.88$ and $K = 150$ the dock stopping road is held only about 23% of the time under reservation (vehicles wait at their last unloading point); about 51% under the reference rule and about 97% under safe release (refused vehicles wait on their current road)
- **Split road:** a vehicle waiting for the crane holds only the 50 m at the dock; under the reference rule at $p = 0.99$, held time falls from 81% for all of road 161 to 65%
- **In-span lifts** (T21, same p): waiting at the dock falls by about a quarter, the plateau does not change: p sets the strength, the in-span share only changes how many vehicles queue on the road

Crane starvation: nominal $\rho < 1$, yet effective capacity falls short

– at $\rho = 0.99$ safe release uses only 74–87% of crane-hours

Okpo Yantai



83% / 87%

Safe release at $\rho = 0.99$, crane-hours used (whole / split): 12–16 points below nominal; erections pile up and the system is unstable

96%

Reservation: cranes close to nominal; vehicles wait at stops for their slot, nothing reaches the network

96%

Reference rule: the cranes lose almost nothing; the plateau falls because the dock queue holds road 161 (81% of the time) and blocks other flows

Starvation = share of time a crane is free while a released erection has not started; under reservation at high ρ about half of the “stuck” time is waiting for the crane slot ($\rho = 0.99$: 53–58%).

■ Whole-route reservation ■ Reference rule (teleport clearing) ■ Safe release — whole road

- - split at stops

Sources: thesis-core a882742, t22_rho.csv and T22 report §2.4 and §3.3. Conditions: saturated demand, $K = 150$, 16 h working day, 8 days, 10 seeds; crane-hours used = time the cranes are held $\div$ time available, dashed diagonal = nominal ρ ; starvation includes the normal lead time of fetching an erection, with the part stuck in the network reported separately (on hover). Okpo rev 1825, Yantai r396.

40–50% of loaded erections were teleported, so the result rests on reservation and safe release; the split road is now built in

① Teleports counted by task (reference rule, whole road, K = 150)

36–47%

Okpo: loaded erections teleported at least once

39–49%

Yantai: the same

39–56%

of these, teleported twice or more (both yards, about half)

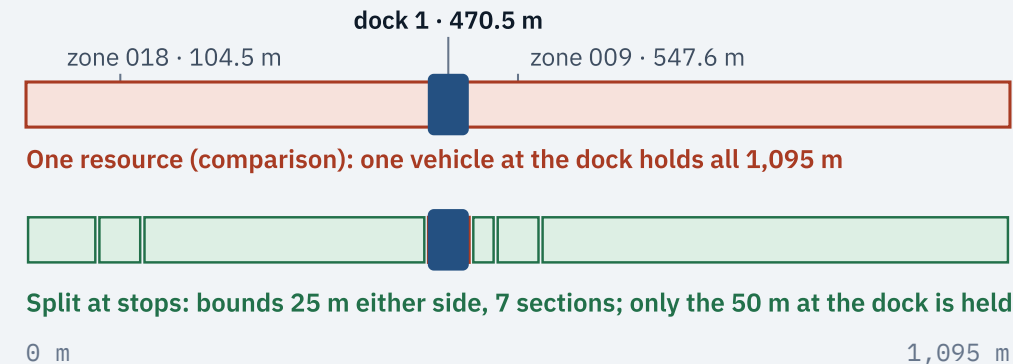
7–11

most teleports of a single task

By construction a vehicle waiting for the crane or unloading at the dock edge is never cleared; those teleported are carrying an erection to the dock, or hold the dock stopping road while waiting for the next section, and each leaves the network for 900 s.

So the reference rule (teleport clearing) is a comparison only; the dock-mouth result rests on whole-route reservation and safe release, neither of which teleports (manifesto §5, rule 2).

② Road 161: 1,095 m long, 9 m wide, three stops along it



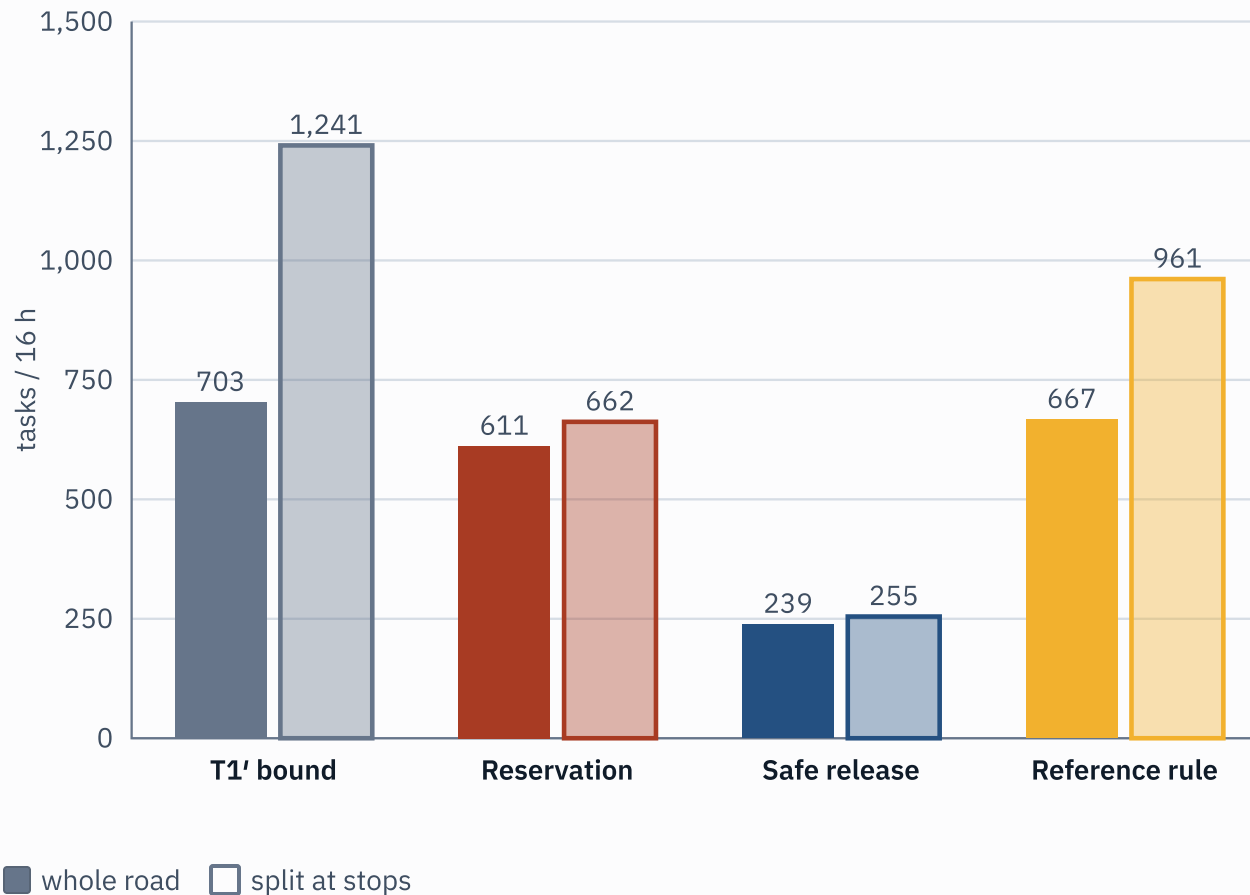
One resource (comparison): one vehicle at the dock holds all 1,095 m

Split at stops: bounds 25 m either side, 7 sections; only the 50 m at the dock is held

T22: the split is a network feature (section = "stops"): roads over 200 m with roadside stops get bounds 25 m either side of each stop. One rule for both yards picks exactly the 4 dock roads; 200 m and 25 m are scenario settings borrowed from railway block sections (UIC 2013) and single-lane mine roadways (Eustace et al. 2021).

The author's imagery check (9 Oct 2026): while a vehicle unloads or waits at the dock edge, neighbouring assembly sites **stay reachable**, so the split is closer to Okpo; the whole road is the comparison.

Erection mix (10 seeds): splitting the dock road moves the capacity interval from [611, 703] to [662, 1,241], and the bottleneck from road to rule

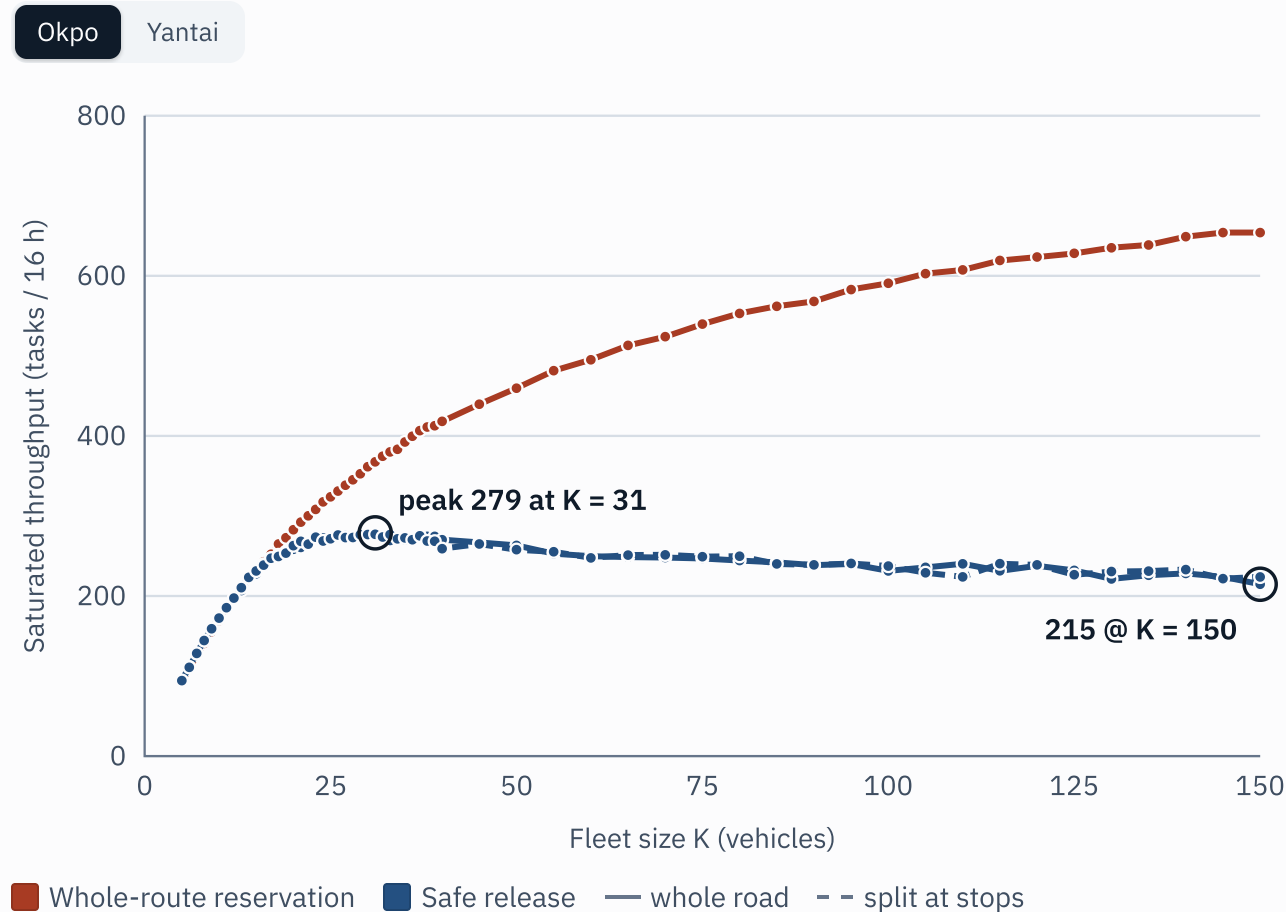


- Erection mix: [611, 703] → split [662, 1,241]; upper edge +76.5%, lower edge +8.3%
 - Plateau / T1': erection mix 0.87 → split 0.53, main scenario 0.34–0.36: change the granularity and the bottleneck moves from road to rule (the C3 / C4 boundary)
 - Okpo main scenario: [627, 1,720] → split [630, 1,857], upper edge only (+8.0%; lower +0.3%); Yantai [455, 1,252]
 - **Bound (T23):** the erection mix's system T1' is 697 (no sampling noise); on the completed mix reservation reaches 86% of the bound, so changing the road still pays
- Interval = [best deadlock-free rule, T1']; of the rules T22 tested, the best is whole-route reservation (above safe release for every $K \geq 20$). The reference rule on the split road has direction locks; its gain still includes teleports (2,394 deadlocks a day at $K = 150$), so it is a comparison only.

The author's probe (9 Oct 2026, 3 seeds) saw this first, in the same direction: T1' 703 → 1,241, reservation +10%; T22 settles it with 10 seeds and the full K grid (reservation at $K = 100$: 585 → 627, +7%).

Sources: thesis-core a882742, t22_interval.csv and T22 report §3.5 and §4 (batches erect_safe a1f485302418, main_safe 656e2da75368 / 930e637e0658; reservation and the reference rule on the whole road reused from T18 and T20 batches); T23 report §0, items 1 and 6. Conditions: erection mix = P6 at 15%, no crane limit; saturated, $K = 5-150$, 4 days, 10 seeds; plateau = mean over $K = 100-150$; 16 h working day; Okpo rev 1825. The main scenario's lower edge 627 is the T20 summary's unrounded 627.45.

H2: safe release, with no teleporting and no deadlock, rises and then falls; reservation, waiting off-road, rises monotonically to a plateau



H2 (manifesto v1): with off-road waiting and a deadlock-free rule, throughput never falls as vehicles are added: a plateau. As long as the extra vehicles occupy network resources, throughput falls beyond K_0 , and K_0 moves right with the holding-area capacity S .

279 → 215

Safe release (whole road): peak at $K = 31 \rightarrow K = 150$ (−23%)

627

Reservation: rises monotonically to its plateau (mean over $K = 100-150$; 654 at $K = 150$)

984

Plateau of the reference rule (teleport clearing, off the chart); highest 1,028 at $K = 150$, with no fall like safe release on the same network

The peak and the fall are a property of the safe-release criterion: about 2/3 of its waits start with a refusal although capacity is free. Without clearing the network jams, sooner with more K (Okpo: 4.3 h at $K = 5$, under 1 h for $K \geq 40$). T27 re-checks the fall; T26 adds a holding area S .

Sources: research manifesto v1 §3, H2 (quoted); thesis-core a882742, T22 report §2.1, §2.2 and §5 and t22_interval.csv; batches main_safe 656e2da75368 (Okpo) and 930e637e0658 (Yantai), with reservation and the reference rule reused from T18's a0ab16fceb94 + 49d2383872e7 and T20's b82aa19b22af. Conditions: main scenario (MY-B flow mix, Roh & Cha 2011), saturated demand, 16 h working day, 4 days, mean of 10 seeds; Okpo rev 1825, Yantai r396.

H2's switches: T22 swept clearing and waiting position, T26 sweeps holding area S; T4 is proved on a corridor first

① Holding area S

idle and waiting vehicles beyond S occupy the roadside; read $K_0(S)$ T26 · planned

② Clearing mode

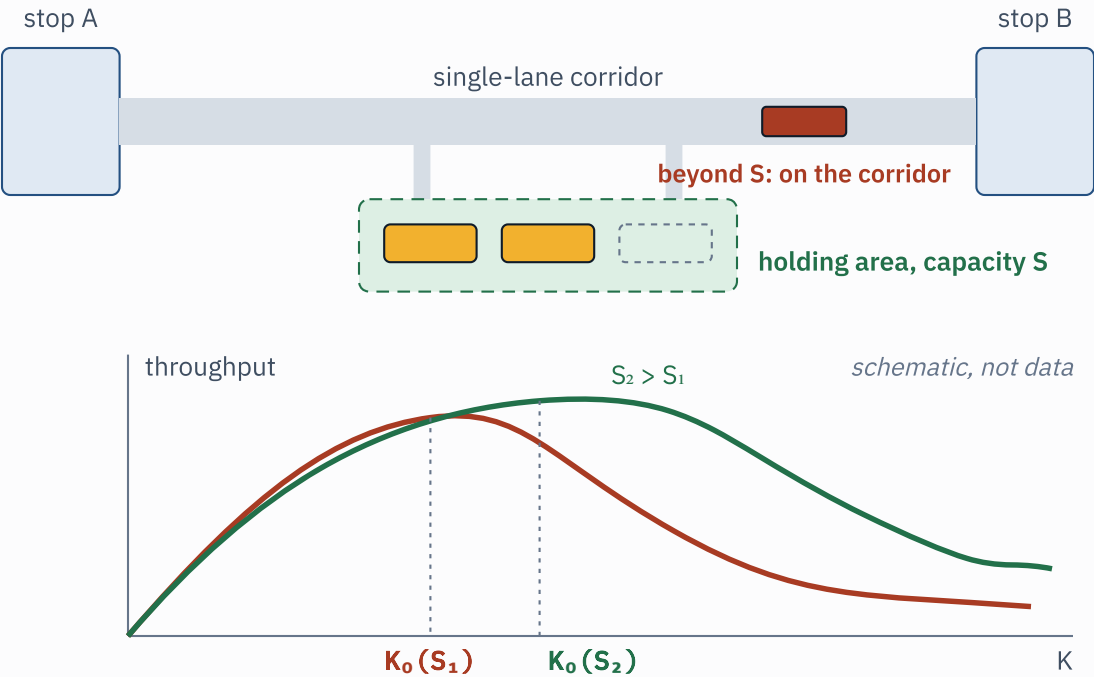
safe release (Banker-type, no vehicle leaves the network): rises then falls; no clearing: the network jams, sooner with more K; teleporting (reference): monotone to a plateau T22 · read

③ Where to wait for the crane

off-road: no fall; on the dock road: falls with p , threshold moving left with K; split: moves right. Holding bays and block length go to T25 T22 · T25

If it holds:

the conditions under which more vehicles hurt, and “how many vehicles a holding area is worth”. If not: a redundancy index, the ratio of marginal throughput to the free-flow margin. Good orchestration can push K_0 back: T27 re-checks the falling branch with a less conservative deadlock-free rule.



T4, non-monotonicity (manifesto §4): on a single-lane corridor with a holding area of capacity S, under greedy release, throughput falls strictly for $K > K_0(S)$. Method: a sample-path construction. Output: chapter 5's design rule “how many vehicles a holding area is worth”.

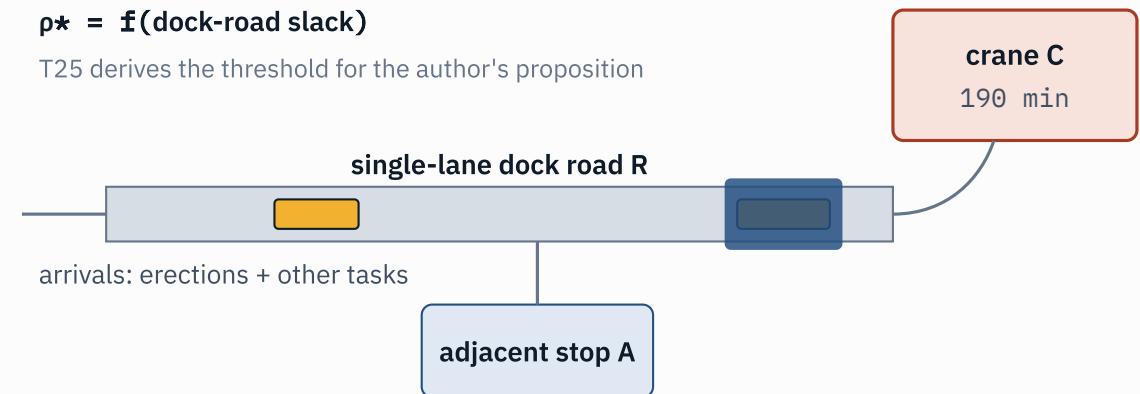
Sources: research manifesto v1 §3, H2 (test, outcomes) and §4, T4; thesis-core a882742, T22 report §2.2, §3.2 and §5; 00_总控/tasks.md T25, T26, T27; the author's review §3, question 3 (three switches, sample-path proof). Safe release mirrors signalmen releasing vehicles on site (KED Global 2024).

Candidate dock-mouth proposition: at nominal load $\rho < 1$, blocking-induced starvation can destabilise the system – T22 reads it without teleporting

- **The channel:** vehicles waiting for the crane block the way to other stops $\rightarrow$ later erections cannot reach the dock and the crane idles because upstream is blocked $\rightarrow$ effective capacity falls below 1, and the system is unstable although nominal $\rho < 1$
- **Restoring stability:** waiting off-road (reservation, a holding bay at the dock) or a dock road used in sections $\rightarrow$ the stable range returns to $\rho < 1$
- **Observable:** the crane starvation rate (read in T22: at $\rho = 0.99$ safe release uses only 74–87% of crane-hours, reservation 96–97%; on Yantai under the reference rule, starvation stuck in the network is twice reservation's)

$$\rho^* = f(\text{dock-road slack})$$

T25 derives the threshold for the author's proposition



Wording: we extend the blocking-induced starvation of queueing networks with blocking to a dock mouth where a crane is in series with an exclusively occupied road. Status: T22 supports the mechanism under rules without teleporting; the author writes the proposition, and the three-resource network (T25) gives the threshold ρ^* .

Theory and position: transferred from queueing networks with blocking; the closest work in the field does not narrow the claim

THEORY SOURCE	WORKS	USE
Queueing networks with blocking	Onvural 1990; Balsamo et al. 2001 checked	foundation
Policy-induced instability at nominal load < 1	Kumar & Seidman 1990; Lu & Kumar 1991; Rybko & Stolyar 1992; Dai 1995 checked	analogy, transfer source; fluid models as the proof tool
Banker-type deadlock avoidance	Habermann 1969 to verify ; Reveliotis 2000	source of safe release; conservative on single lanes in general

“Nominal load < 1 at each station is not sufficient for stability” is best quoted from Dai 1995's abstract; Dai's model has no blocking and infinite buffers, a difference stated when transferring it to road occupancy.

	TAO ET AL. 2017	THIS STUDY
Road occupancy	capacity counted in vehicles (2 on main roads, 1 on side roads)	exclusive use and blocking time that change with the load
Rules, deadlock	one (first come, first through), deadlock not handled	three rules; reservation and safe release are deadlock-free
Road-level output	none (no count, duration or road utilisation of blocking reported)	thresholds, dock road held, crane starvation
Bound	none (simulation only)	T1', T1'' and the capacity interval
Yards	one	two (more in chapter 6)

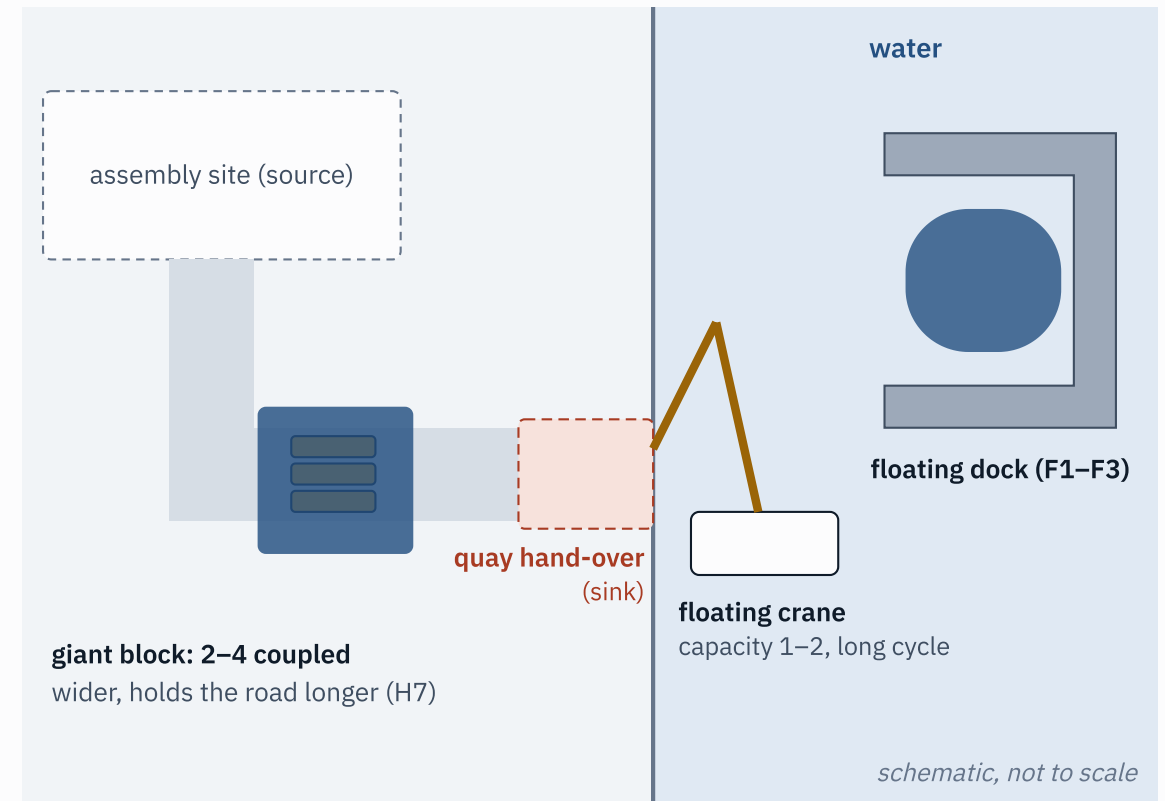
What is new: ZhangSG2025 schedules gantry cranes and transporters separately; here crane cadence and road-owning loaded vehicles share one simulation and one bound. Within the searches of the review's appendix B and T24, this is the first work to read shipyard throughput with waiting position, control granularity, fleet size and crane utilisation as joint variables.

Sources: T24 notes (00..小论文/Reference/notes/occupancy_anchors_20261009.md) §3.1 (item-by-item comparison with Tao et al. 2017, full text, T24b) and §4.1, §4.2, §4.4 (bibliographic checks and quotations of transfer sources); the author's review, appendix B; thesis-core a882742, T22 report §9; T20 and T21 reports §7 (comparison with ZhangSG2025). Transfer sources serve as theoretical support, not as prior claims.

Floating docks: a quay-side sink plus a floating crane; giant blocks on coupled transporters are H7's case

- **Three floating docks:** Hanwha Ocean IR 4Q24 p. 29 and 2Q26 p. 27, “Floating Dock : 3EA”; on the imagery (rev 1833) F1 $\approx 457 \times 82$ m, F2 $\approx 360 \times 77$ m, F3 $\approx 472 \times 80$ m, container ships in F1 and F3
- **Fits the base scenario:** LNG carriers in dry docks 1 and 2 (author; four at once in dock 1, Seoul Shinmun 2024, Hanwha 2025); all 33 merchant ships there would give $p = 1.29\text{--}1.60$
- **Model:** Okpo lifts with floating cranes (3,600 t class, two in tandem for big blocks): vehicles bring giant blocks to a shore site in reach (the sink); the floating crane is a resource of capacity 1–2 with a long cycle, reusing the crane code; roll-on only as comparison
- **Giant blocks:** over 500 t, sometimes 4 coupled transporters; “about 10 giant blocks a ship” is from 고시근 et al. 2011 and Samsung, not Okpo; wider, holding the road longer, returning as a team $\rightarrow$ H7 and MY-B's coupled carriage

T26 follows T27 and T25; T27's step zero moves Okpo to rev 1833 (20_数据/船厂地图). Each dock's size and the speed of coupled transporters have no public source yet: scenario settings, reported side by side.



Established (T20–T24, all reviewed): each feeds a thesis figure or section

ITEM	STATUS	OUTPUT (WHERE IN THE THESIS)
Cranes as dock resources, erections at crane cadence, in-span lifts; the crane row and window edge term of T1' (draft proof exists; the author rewrites it)	established · T20–T21	Chapter 3 methods and parameter table
T22: safe release (exact, deadlock-free), no-clearing switch; split road; crane starvation, teleport diagnosis; p curve and threshold at each K; capacity intervals; H2's rise and fall	established · T22	Chapter 3: dock-section mechanism figure and threshold table; interval section; H2 section
T23: system T1' and route-free T1'' (draft proof, the author rewrites); route slack in the bound (through corridor +45% under crane cadence, dock stopping roads +4–8%)	established · T23	Chapter 3 bound section; dual-price column of chapter 5's rate table
T24: transfer sources checked; public sources on floating docks and giant blocks; Tao et al. 2017 compared in full	established · T24	References; the introduction's comparison; floating-dock parameters

Running and queued: T27 first, then T25 and T26; each names the figure or section it produces

ITEM	STATUS	OUTPUT (WHERE IN THE THESIS)
T27: a less conservative deadlock-free rule (Banker plus step-wise feasibility search, or claims cut at the next safe holding point, T5), with the refusal share. Step zero: 20_数据/船厂地图 (Okpo rev 1833), direction locks at stop U-turns, reservation waits split into crane / road	running	Chapter 4 rule comparison; re-check of the lower edge, H2's decline and H5's cost of safety
T25: three-resource dock network (p^* against dock-road slack); upgrade simulations; block-length curve; door granularity	queued · after T27	The proposition's threshold; the first road–vehicle exchange-rate table (chapter 5)
T26: floating-dock scenario; finite holding area S and $K_0(S)$	queued · after T25	H2's $K_0(S)$ curve and “how many vehicles a holding area is worth”; H7's floating-dock case

For the author once T27 is done: the wording of H1 part 3 (readings on slide 8); the new numbers of H1 part 2 (vehicles needed at yard-wide volume); the main basis (16 h or 24 h working day, daily mean or peak).

Risk

Safe release is conservative (2/3 of its waits start with a refusal), so its thresholds and H2 numbers will move under a looser rule; the split's 200 m and 25 m are scenario settings.

Mitigation

T27's rule enters through the same batches; the result leads with the qualitative relations, numbers carrying rule and K; both granularities are reported and T25's block-length sweep turns 25 m into a curve.

Sources: 00_总控/tasks.md T25–T27 (10 Oct 2026; T27 brief runs/2026-10-10_T27_prompt.md); T22 report §2.1, §10 and §15 (risks and how to strengthen; items for the author); reviews/T22.md §5. T25–T27 have no results yet; they are listed as plans.

Target journals: the dock mouth goes into paper one (chapters 2–3); B-class first, RESS (A) as the stretch

First choice (B)

Transportmetrica A: Transport Science class B

Ocean Engineering class B

The capacity interval, p thresholds and waiting position are transport science; the dock mouth and erection are shipyard mechanisms.

Stretch (A)

Reliability Engineering & System Safety class A

Condition: the proposition is proved as a stability / resilience result, “blocking-induced starvation, instability and recovery”. This sub-study fits it best: crane starvation observes the instability; off-road waiting and splitting are the levers of recovery.

Alternatives (B)

IEEE Transactions on Automation Science and Engineering class B
Applied Mathematical Modelling class B

An automation-systems framing of safe release and sectioned control; a modelling framing of the bound and the three-resource network.

Supplementary (C)

Computers & Industrial Engineering class C
Journal of Ship Production and Design class C

Engineering outputs: design rules on holding bays, sectioned dock roads and floating docks.

B-class first; A-class as the stretch, condition on its card. The school requires at least one journal paper of class B or above; MY-A is already published in Ocean Engineering (B), and MY-B targets Ocean Engineering.

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To close: the dock-mouth section answers “where to wait, how to split the road, and how many vehicles wait on it”

1

The crane is in the model: erections at crane cadence, in- and out-of-span from the map, and one parameter p places a yard on the curve.

2

The mechanism is read (T22, 10 seeds): off-road waiting does not fall; rules that wait on the road, safe release without teleporting included, fall with p , with the threshold moving left with K and right with splitting; cranes can starve at nominal $p < 1$.

3

Next: T27 re-checks the lower edge and H2's decline; T25's three-resource network gives the threshold p^* for the author's proposition; together they become chapter 3's dock section and chapter 5's design rules.

[← Back to the thesis briefing](#)[Capacity interval →](#)[Traffic orchestration →](#)[Fleet-network co-design →](#)[Fleet sizing \(MY-B\): coupled carriage →](#)

Thank you.